Eberhard Karls Universität Tübingen

Faculty of Economics and Social Sciences

Chair of Statistics, Econometrics and Quantitative Methods

Chair of Statistics, Econometrics and Empirical Economics

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Google Search Indicators and Nowcasting German Unemployment: Early vs. Late Information

Author:	Alexander Unger
Matriculation No.:	6946672
E-mail:	youremail@stud.uni-tuebingen.de
Supervisor:	Alexander Reining
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Abstract

This paper evaluates whether Google search data can improve real-time nowcasts of German unemployment. Following Zimmerman's biweekly aggregation framework, weekly search indicators for 2011–2023 are transformed into early-month (W12) and previous-month (W34) components to reflect the timing of the German unemployment measurement process. After confirming that both unemployment and search series are nonstationary and not cointegrated, the analysis is conducted in first differences using rolling-window AR and ARX benchmarks as well as mixed-frequency MIDAS regressions. The study compares whether biweekly aggregation removes relevant high-frequency information by contrasting traditional W12/W34 models with unrestricted MIDAS specifications that use weekly signals directly. Forecast accuracy over 2019–2023, including the COVID-19 shock, is assessed using RMSE, MAE, and Diebold–Mariano tests. Results show that German unemployment is highly persistent, with limited incremental predictive power from search indicators. While weekly MIDAS models capture slightly more high-frequency variation, they do not substantially outperform simple AR benchmarks. These findings suggest that the informational content of Google search data for German unemployment is modest, and sensitive to the chosen temporal aggregation.

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1 Introduction

Timely and reliable macroeconomic indicators are essential for economic research institutions and policymakers, particularly central banks, as they monitor current economic conditions and assess the stance of fiscal and monetary policy. However, in many countries key statistics, most notably unemployment, are published only with considerable delay after the reference period. Such publication lags create informational gaps that can impede effective decision-making, especially during periods of heightened uncertainty or rapid economic change. Episodes such as the COVID-19 pandemic or the 2008 financial crisis demonstrated how delays in labor-market and consumption indicators complicated the ability of authorities to respond swiftly to emerging developments.

These challenges stem largely from the nature of traditional data collection. Survey-based and administrative statistics are often resource-intensive, require extensive processing, and therefore cannot capture economic dynamics in real time. As a result, official indicators may fail to reflect evolving conditions precisely when timely insights are most valuable.

Against this backdrop, researchers and institutions have increasingly turned to high-frequency, unconventional data sources to complement official statistics and improve short-term forecasting. Digital traces such as job-posting data, mobility patterns, and, most prominently, search-engine activity, offer new possibilities for real-time monitoring. These data sources reflect aspects of individual behavior and expectations at a much higher frequency than traditional statistics.

Among these alternatives, search-engine queries stand out due to their ubiquity and their strong link to information-seeking behavior. Search engines constitute the primary interface through which users access online information, and they are used across virtually all demographic and socioeconomic groups for purposes ranging from job search to product research and education. Consequently, search data provide large-scale behavioral signals that reveal what individuals are currently concerned with or intend to do. Google's dominant market position ensures that Google Trends offers broad and demographically diverse coverage of search behavior.

The theoretical appeal of search data lies in their behavioral immediacy. Individuals typically conduct online searches before engaging in economic actions, such as applying for jobs or adjusting consumption behavior. For example, rising search volumes for terms like "job center," "layoff," or "unemployment benefits" may indicate emerging stress in the labor market before it becomes visible in official statistics. When aggregated across millions of users, such search activities form a micro-level digital footprint that can serve as a high-frequency proxy for macroeconomic conditions. In this sense, search data operate as a near real-time indicator of economic sentiment and intention, making them particularly suited for nowcasting applications where official data are published with a delay.

The idea that digital search behavior captures real-time shifts in economic sentiment has inspired a series of empirical studies seeking to quantify its predictive value. Early contributions in this area laid the groundwork for using online search data as an economic indicator.

One of the earliest contributions to this line of research comes from Ettredge et al. (2005) who examined whether search volumes from WordTracker's metasearch engine could anticipate U.S. unemployment. Although their short-term signals were noisy, they showed that smoothed, longer-run trends in job-related searches were strongly associated with future unemployment, in some cases outperforming weekly claims. This early work established the core intuition that aggregated online search activity captures evolving labor-market expectations before they appear in official data releases.

Building on this foundation, Zimmermann (2009) introduced the "Google Jobs Index," the first systematic attempt to quantify the informational content of Google search queries for macroeconomic forecasting. Using German data from 2004–2009, they demonstrated that job-related searches reliably lead unemployment releases by several weeks. Their results established Google Trends (GT) data as a practical, publicly available high-frequency proxy for labor-market stress.

A major conceptual advance came from CHOI and VARIAN (2012), who extended the use of GT data beyond unemployment to a broad set of macroeconomic indicators. Rather than focusing on multi-step forecasting, they emphasized nowcasting, using contemporaneous search information to fill in publication delays. Their autoregressive models augmented with search volumes (e.g., "unemployment benefits," "car sales") consistently improved out-of-sample performance, especially around turning points. This work positioned Google Trends as a tool for real-time macroeconomic monitoring.

Subsequent research tested the robustness of this approach across different economic environments. For instance, CARRIÈRE-SWALLOW and LABBÉ (2010) showed that search-based indices can predict automobile sales in Chile, improving both in-sample and out-of-sample accuracy while capturing market reversals earlier than official statistics. Their findings reinforced the notion that search activity does not merely respond to economic outcomes but contains leading information.

Further methodological refinement came from D'Amuri and Marcucci (2010), who evaluated whether a Google Job Search Index improves U.S. unemployment forecasts relative to traditional indicators. Comparing more than 500 ARMA and ARMAX specifications and conducting extensive forecast evaluation (Diebold–Mariano tests, forecast encompassing, Reality Check), they found that search-augmented models outperformed traditional benchmarks in the majority of U.S. states. This study established search data as a statistically robust leading indicator.

At the same time, research broadened beyond labor markets. Goel et al. (2010) applied search data from Yahoo! to predict consumer demand for entertainment products. They showed that search intensity strongly correlates with future sales and may surpass traditional predictors in rapidly evolving markets. This work underscored that the predictive value of search data is context-dependent and particularly strong when official data are sparse or released with delay.

More recently, studies have increasingly combined high-frequency search data with mixed-frequency econometric techniques, enabling real-time estimation of monthly figures using daily or weekly information. For instance, Costa et al. (2024) propose a MIDAS framework for nowcasting Portuguese unemployment using daily GT series, overcoming historical GT sampling constraints through overlapping-window collection and temporal disaggregation. Their results show that daily GT predictors substantially outperform both monthly GT predictors and benchmark models, especially during economically volatile periods such as COVID-19. Their methodological innovation, linking real-time availability, high-frequency GT data, and mixed-frequency methods, which represents the most advanced stage of search-based nowcasting.

Despite a large international literature on search-based economic forecasting, relatively few recent studies re-examine the German labour market using up-to-date Google Trends data, particularly during recessionary periods such as the COVID-19 shock. This gap is notable given Germany's economic importance and the increasing prevalence of job-related online search behaviour.

Building on the biweekly aggregation framework introduced by Zimmermann (2009), this seminar paper investigates whether Google search intensity improves the real-time nowcasting of German unemployment. To assess this potential, I compile a monthly panel for 2011–2023 that combines official unemployment figures with biweekly Google Trends indicators representing the two windows used in Zimmermann (2009) approach: an early-month component (W12) and a previous-month component (W34). A set of benchmark nowcasting models is estimated in first differences, including an AR(1) and an ARX specification incorporating contemporaneous biweekly search intensity. These models serve as a baseline for evaluating whether search data provide incremental predictive content beyond unemployment persistence alone.

A key contribution of this paper is to investigate whether the biweekly aggregation of search data, standard in earlier work, leads to a loss of informational content. To do so, I complement the biweekly ARX models with unrestricted MIDAS regressions that directly incorporate the underlying weekly search data. This mixed-frequency approach allows me to test whether weekly information improves unemployment nowcasts relative to their biweekly counterparts. All models are evaluated using a rolling real-time nowcasting design over 2019–2023, with forecast accuracy assessed through RMSE, MAE, and Diebold–Mariano tests. The empirical results provide new evidence on the usefulness and limitations, of Google search indicators for monitoring short-run labour-market dynamics in Germany.

2 Data

2.1 Official unemployment series (Germany)

The target variable of this seminar is the monthly, seasonally unadjusted unemployment rate published by the *Bundesagentur für Arbeit*. Following Zimmermann (2009) , I use the *Arbeitslosenquote bezogen auf alle zivilen Erwerbspersonen*, which measures unemployment relative to the civilian labour force and is reported in percentage points.

Using the unadjusted series is deliberate. Seasonal patterns such as winter layoffs, school-to-work transitions or holiday effects carry information that may be reflected in high-frequency search activity. Removing these variations ex ante would risk discarding precisely the fluctuations Google Trends may capture. In the empirical specification, seasonality is therefore controlled for explicitly through monthly dummy variables rather than through pre-adjustment.

A key institutional feature is the timing of unemployment measurement and publication. Although figures are released at the end of month M, they refer to labour-market conditions up to the middle of that month. This reporting lag creates a natural real-time forecasting environment: at the time a forecaster forms a nowcast for month M, the latest available official observation refers to month $M - 1$. Search data from different parts of the month may therefore contain distinct information sets depending on whether they precede or follow the publication of the previous unemployment number.

The sample used in this study covers mid-2011 to January 2023. The starting date is chosen for both statistical and economic reasons. On the data side, early Google search data were collected via the now-discontinued Google Insights interface, and several methodological revisions before 2010 make the pre-2011 period difficult to compare consistently with modern Google Trends indices. Internet penetration also increased substantially during the 2000s, implying that early search volumes may not yet reflect broad population behaviour. Beginning in 2011 ensures that search activity is measured within a stable and fully digital environment.

From an economic perspective, the mid-2010s mark a phase in which the structural labour-market adjustments associated with the Hartz reforms had already materialized. Starting the sample after this transition avoids large structural breaks and yields a more homogeneous relationship between job-search behaviour and unemployment dynamics. The chosen window also includes distinct macroeconomic phases like post–Eurozone crisis recovery, pre-pandemic lows, the COVID-19 shock, and the inflationary period after 2022. Those offer a rich environment for evaluating real-time predictive performance.

While excluding the earlier 2004–2009 period prevents an exact replication of the original sample in Zimmermann (2009), the improved data quality and economic comparability of the 2011–2023 period make it more suitable for the nowcasting analysis in this seminar.

2.2 Google trends data

Google Trends provides anonymized indicators of search activity by reporting, for each keyword and region, a 0–100 index of relative search intensity. The index does not reflect absolute search volumes; instead, it is normalized such that 100 corresponds to the peak share of that keyword within the selected time window, and all other values express proportions of that peak. As a consequence, the data measure relative interest rather than levels, and series for different keywords are not comparable in levels unless transformed or standardized.

A second implication is that Google internally draws daily samples from the full query stream. Repeating the same download on different days can therefore yield slightly different values (CARRIÈRE-SWALLOW and LABBÉ, 2010). While some studies mitigate this by averaging multiple repeated downloads, I rely on single downloads due to time constraints; this affects only the noise level, not the structure of the data.

Because the index is normalized within each query and window, Trends data must be preprocessed before they can be used in time-series models that require consistent scaling across long samples, especially the models used in this study such as ARX or unrestricted MIDAS regressions.

2.3 Constructing of a consistent weekly Google Trends History

Google Trends provides weekly data only for horizons of about five years. To obtain a continuous weekly series for 2011–2023, I follow the standard "stitching" approach from the literature: multiple overlapping windows are downloaded, and the later windows are rescaled to match the earlier ones in their overlap. Specifically, for any two adjacent windows j and $j+1$, a scaling factor is computed from the ratio of their overlapping weekly values, and the later window is multiplied by this factor before concatenation.

$$s_{\,j+1\leftarrow j} \;=\; \operatorname{median}\left\{\, \frac{x_t^{(j)}}{x_t^{(j+1)}} \,\right\}. \tag{1}$$

This procedure reconciles the window-specific 0–100 normalizations and yields a coherent weekly series for each keyword. The resulting data remain relative indices, but they are now internally consistent over time, which is essential for the MIDAS models that use the weekly information directly.

Detailed robustness checks (e.g., alternative scaling ratios or trimming) affect only small-value weeks and do not alter the empirical results; therefore, I keep the description concise here.

2.4 Biweekly aggregation approach

For the baseline replication of Zimmermann (2009), I construct biweekly Google Trends indicators that mirror the timing structure of German unemployment measurement. The Federal Employment Agency publishes the unemployment rate for month M at the end of that month, although the reference date lies around the middle of M. Search behaviour may therefore reflect (i) earlier information about economic conditions in month M or (ii) later signals that coincide with the measurement of unemployment.

A key motivation for the biweekly split originates from Zimmermann (2009). The unemployment rate for month $M-1$ is released at the *end of month* $M-1$, i.e. precisely between the two biweekly intervals. As a result, search activity in the first half of month M ($W12_M$) may react to the publication of U_{M-1},for example, due to media coverage or individual responses to newly announced labour-market figures. By contrast, search activity in the second half of month $M-1$ ($W34_{M-1}$) is affected only by the earlier release of U_{M-2}, which is less likely to influence behaviour two weeks later. In this sense, $W34_{M-1}$ is expected to contain "cleaner" and more forward-looking information about the labour market than $W12_M$, which may be contaminated by news effects. To approximate these two information sets for a given reference month M, I build two aggregates:

- $\mathbf{W12}_M$: average weekly search activity in the first half of month M (days 1–15). This window occurs *after* the publication of unemployment for $M-1$ and overlaps with the measurement date for month M.

- $\mathbf{W34}_{M-1}$: average weekly search activity in the second half of month $M-1$ (days 16–end of month), aligned forward to month M. This window captures *earlier* search signals that precede the measurement of unemployment in month M and are not influenced by the release of U_{M-1}.

Unlike Zimmermann (2009), who proportionally split weekly data at intra-week boundaries, I assign each weekly observation to the half-month containing its Sunday timestamp. A concrete calendar illustration (without the three-Sundays edge case) is provided in Appendix 9.1.

The resulting biweekly panel spans June 2011 to January 2023, yielding $T = 140$ monthly observations. This horizon provides sufficient degrees of freedom for time-series estimation while allowing a clear separation between *earlier* and *later* search signals relative to each reference month.

This setup allows me to address two central questions. First, whether search intensity during $\mathrm{W12}_M$ contains more predictive information for changes in unemployment than the earlier $\mathrm{W34}_{M-1}$ component, i.e. whether "later" search signals are more informative than "earlier" ones. Second, in the subsequent unrestricted MIDAS models, I use the full weekly series directly to evaluate whether biweekly averaging discards relevant high-frequency information and whether weekly MIDAS specifications can improve nowcasting performance relative to traditional biweekly aggregation.

2.5 The selection of google trends variables

The predictive value of search data depends critically on the choice of keywords. Google Trends reflects behavioural intentions rather than economic quantities, and unsuitable keywords can introduce news-driven noise or capture unrelated dynamics. For this reason it is important to carefully select the variables in order to extract signals that plausibly relate to labour-market flows.

In the original study of Zimmermann (2009), search behaviour was grouped into four conceptual categories: (k1) contacts with the unemployment office, (k2) interest in the unemployment rate itself, (k3) activity around personnel consulting, and (k4) searches related to job boards. As the structure of the online job search changed markedly after 2010, with especially the emergence of dominant private platforms, I select a smaller, more coherent set of modern search terms that align with the original k1 and k4 concepts while excluding news-driven or unstable categories.

The term arbeitsamt, although officially replaced by Bundesagentur für Arbeit in 2004, remains a colloquial, widely used search term associated with unemployment services. I believe it captures information seeking related to registration, benefits, or interactions with the public employment service and thus serves as a stable proxy for impending or actual inflows into unemployment. It corresponds closely to the original k1 category of Zimmermann (2009).

The keyword *Jobbörse* refers to the official online job portal of the Federal Employment Agency. Search activity for this term reflects use of a central institutional job-search channel that is tightly connected to unemployment services. This word is part of the k4 job-board basket but represents a conceptually cleaner and more stable public-platform signal over the 2011–2023 period.

The terms stepstone and indeed represent the dominant private job search platforms in Germany after 2010. Including them captures a substantial share of modern job-search behaviour that was not yet prevalent during the 2004–2009 sample used by Zimmermann (2009). These keywords generalise the original k4 category to a more current set of platforms while avoiding brand drift in earlier job-board names that have since lost relevance.

The keyword bewerbung (job application) encompasses searches related to writing cover letters, CV preparation, and the application process as a whole. It reflects a broader job-search intention that is conceptually related to the employment transition margin and is behaviourally analogous to Zimmermann (2009)'s k3 category.

2.6 Restrictions

Before turning to the methology, it is important to acknowledge several limitations of Google Trends data that affect interpretation and forecasting accuracy.

As mentioned Google Trends is based on daily subsamples of the full query stream, which implies that identical requests on different days can yield slightly different series (CARRIÈRE-SWALLOW and LABBÉ, 2010). This introduces measurement noise that can affect high-frequency models such as unrestricted MIDAS. Google also applies an internal thresholding rule for low-volume queries; although the exact cutoff is unknown, it may suppress or smooth observations in periods of weak search intensity. Both aspects imply that Trends data should be interpreted as noisy behavioural proxies rather than precise measurements.

Moreover, search terms may carry multiple meanings or evolve over time. Although the selected keywords in this study are behaviourally well-defined, they may still capture heterogeneous intentions (e.g. "bewerbung" searches by students, career changers, or unemployed individuals). Google's query classification system is not observable to the researcher, and keyword-based indices cannot distinguish between different user groups.

Google Trends does not provide demographic or labour-force identifiers due to privacy constraints. As emphasised in D'Amuri and Marcucci (2010), search activity reflects the behaviour of both unemployed and employed individuals. Because unemployed job search tends to be countercyclical, while on-the-job search is typically procyclical, the resulting aggregate index may blur the distinction between different labour-market states. This limits the ability to interpret changes in search intensity as direct measures of unemployment inflows or outflows.

Furtheremore, weekly data do not align perfectly with half-month intervals, so some months contain uneven numbers of observations in W12 or W34 segments. Given the twelve-year sample and the averaging structure, such irregularities are expected to average out, but they nevertheless introduce small timing mismatches in the biweekly indicators.

Overall these limitations imply that Google Trends indicators should be interpreted as timely but noisy behavioural signals. Their usefulness for nowcasting depends on whether the underlying predictive component is strong enough to dominate sampling error, semantic ambiguity, and variation in user composition.

3 Deycriptive statistics and Visual Inspection

3.1 Descriptives

In this section, I take a structured look at the biweekly search indicators in order to (i) verify that the W12–W34prev aggregation behaves coherently over time and (ii) document broad comovement patterns between the search indicators and unemployment. Figure 1 displays both components for each keyword.

Across all series, the most notable feature is the strong and highly regular seasonality. Public-sector related terms such as arbeitsamt, jobbörse and bewerbung exhibit clear intra-year cycles with pronounced peaks at the turn of the year and troughs in summer months. A second consistent pattern is that the W34prev component tends to show sharper and more regular seasonal spikes than the corresponding W12 values. This is particularly evident for arbeitsamt and bewerbung. This asymmetry aligns with the timing structure of the biweekly aggregation: W34prev captures search behaviour before the unemployment figure for month $M-1$ is released and is therefore unaffected by announcement effects, whereas W12 may partly reflect reactions to the publication of U_{M-1}.

This distinction between the two biweekly components is also clearly visible in the seasonal boxplots shown in Figures 2 and 3. For the W34prev indicators, the January spike is particularly pronounced and occurs consistently across all keywords, followed by a sharp decline into February. This pattern is far less pronounced in the W12 components, where month-to-month differences are visibly flatter.

Long-run patterns also differ systematically across keywords. Searches for arbeitsamt and jobbörse display a persistent downward trend over 2011–2023, consistent with declining usage of public employment-service channels. By contrast, private job portals such as indeed show a pronounced upward trajectory during the 2010s, reflecting structural digitalisation and the increased importance of private platforms in German job search. The keyword bewerbung exhibits a smooth and monotonic decline, consistent with reduced reliance on generic application-related searches in favour of platform-integrated application processes.

Several series also feature episodic bursts, most notably around 2020–2021 during the COVID-19 shock. Spikes in arbeitsamt, indeed and stepstone suggest heightened job-search uncertainty and sudden shifts in labour-market activity during the pandemic. In contrast, jobbörse and bewerbung appear less affected, consistent with their declining long-term usage.

Overall the biweekly aggregation looks internally coherent and that both series track similar underlying behavioural patterns, with W34prev often showing a clearer seasonal structure.

4 Stationarity, differencing, and implications for model choice

Before estimating the forecasting models, it is necessary to examine the time-series properties of the unemployment rate and the biweekly Google Trends indicators. Whether a variable is stationary or integrated determines whether the models should be estimated in levels or in differences. It is worth noting that Zimmermann (2009) do not conduct any formal stationarity their error–correction framework. Their empirical model is estimated directly in seasonal differences without verifying the integration order of unemployment or the search indicators. For correct modeling purposes it is essential to investigate these properties explicitly in the present study.

I therefore apply Augmented Dickey–Fuller (ADF) tests to all series. The ADF test extends the classical Dickey–Fuller regression by adding lagged differences to control for serial correlation in the residuals. For each series, lags are selected automatically by the AIC criterion with an upper bound of 13, and critical values follow MacKinnon (2010) as implemented in `statsmodels`.

Because the ADF test is sensitive to the choice of deterministic terms, I select the appropriate specification (no constant, constant, or constant and trend) based on an inspection of the plotted series. Series that exhibit a clear trend are tested under the trend specification, while those fluctuating around a nonzero mean are tested with a constant. This approach ensures adequate power without overfitting deterministic components.

The ADF regression with p augmenting as shown in table 17.3 in Hamilton (1994) is

$$y_t = \zeta_1 \Delta y_{t-1} + \zeta_2 \Delta y_{t-2} + \cdots + \zeta_{p-1} \Delta y_{t-p+1} + \alpha y_{t-p} + \beta \delta_t + \epsilon_t \tag{2}$$

Table 2 reports the ADF results for the variables in levels. Across all Google Trends indicators and for the unemployment rate, the null hypothesis of a unit root cannot be rejected at the 5% significance level, indicating that the series are nonstationary in levels. To confirm whether the variables are integrated of order one, I repeat the tests for the first differences using the no-constant specification, appropriate for series that fluctuate around a zero mean. As shown in Table 3, the ADF statistics for the first-differenced series strongly reject the unit-root null for all variables. Taken together, the results imply that all variables are $I(1)$ and become stationary after first differencing.

For the usage of high frequency models such as the MIDAS specification mdoels, the variables enter at their original weekly frequency rather than in biweekly averages. For this reason, I also examine the stationarity properties of the weekly Google Trends indicators. The ADF results reported in Table 4 show that the weekly stepstone and indeed series remain nonstationary in levels, whereas the weekly arbeitsamt, jobbörse, and bewerbung series reject the unit-root null under the trend specification. In contrast, all weekly series become unambiguously stationary after first differencing. These findings confirm that the weekly indicators can either be used directly in MIDAS in levels (when trending but stationary around a deterministic trend) or entered in first differences when strict stationarity is required.

Overall, the stationarity analysis establishes a clear pattern for the modelling strategy adopted in the remainder of the paper. For the biweekly ARX models, all Google Trends indicators and unemployment are treated as $I(1)$ variables and therefore enter in first differences unless cointegration can be shown to hold. For the unrestricted MIDAS regressions, the weekly indicators provide a richer frequency structure: trending weekly series such as jobbörse, arebeitsamt and bewerbung can be included in levels as long as deterministic components are controlled for. These properties ensure that the subsequent forecasting models rest on well-defined time-series behaviour and avoid spurious regressions arising from nonstationary combinations of unrelated series.

4.1 A notion on cointegration

Since the unemployment rate and the biweekly Google Trends indicators are found to be $I(1)$, one possibility would be to estimate an Error Correction Model (ECM) if the variables share a stable long-run relationship. In this context it is important to note that the ECM specification used by Zimmermann (2009) is not derived from a formal cointegration analysis (Engle and Granger, 1987). Neither study performs an Engle–Granger nor a Johansen test, and the "error-correction term" in their model is implicitly taken to be the lagged level of unemployment rather than the residual from a cointegrating regression. Consequently, the specification resembles an autoregressive model in differences with seasonal controls rather than a structurally motivated ECM.
To assess whether a genuine long-run relationship exists in the present data, I test for pairwise cointegration between unemployment and each biweekly Google Trends indicator using the Engle–Granger procedure. In the first step, unemployment is regressed on each search indicator in levels (including a constant and seasonal dummies), and the residuals are extracted. Under cointegration, these residuals should be stationary. The appropriate critical values are those of Case 2 in Hamilton (1994), since the long-run regression includes a constant but no trend.
Before turning to the test statistics, it is useful to note that several Google Trends series, most notably *indeed* and *stepstone*, exhibit pronounced upward trends that are not mirrored in the unemployment series. From an economic perspective, this already weakens the case for a stable long-run equilibrium relationship. The formal Engle–Granger tests confirm this intuition. Table 6 reports the Engle–Granger ADF statistics for the residuals of the long-run regressions between unemployment and each biweekly Google Trends indicator. Using the appropriate Case 2 critical values from Hamilton (1994), the unit-root null is rejected at the 5% level only for two series (stepstone_W12 and indeed_W34prev), while all remaining eight indicators clearly fail to reject the null. Even for the two borderline cases, the test statistics lie only slightly below the 5% threshold, and the rejection is unstable across deterministic specifications.
The statistical evidence is further undermined by the visual behaviour of the Engle–Granger residuals (Figure 4). Rather than exhibiting stationary mean-reverting dynamics,

the residuals display long, persistent drifts, seasonal patterns, structural breaks, and pronounced deviations during episodes such as the COVID–19 crisis. Such behaviour is inconsistent with the notion of a stable equilibrium error term and therefore contradicts the existence of a meaningful cointegrating relationship.

Taken together, these results imply that the unemployment rate and the biweekly search indicators are not cointegrated. In the absence of cointegration, an ECM is not appropriate, and following the Granger Representation Theorem, the correct empirical specification for non-cointegrated $I(1)$ variables is an ARX model in first differences. This avoids spurious regression behaviour and ensures that the short-run dynamic effects of search activity on unemployment are identified without imposing an unfounded long-run equilibrium structure.

Cointegration is not assessed for the weekly Google Trends indicators used in the subsequent MIDAS models, because the MIDAS framework is designed to capture short-run mixed-frequency dynamics rather than long-run equilibria. Weekly variables therefore enter the MIDAS regressions either in levels (when adequately detrended) or in first differences, depending on their stationarity properties.

5 Econometric Models

The empirical analysis adopts a nowcasting perspective. Nowcasting refers to the real-time prediction of the present or very recent past, i.e. producing an estimate of the unemployment rate for month M before the official figure is released. At that point, the forecaster observes unemployment only up to $M-1$, while weekly Google search indicators are already available within month M. The econometric task is therefore to map past unemployment and high-frequency search information into a real-time estimate of U_M.

To address this task, I consider three classes of models. As a benchmark, unemployment is modelled as a univariate autoregressive process in first differences. Second, ARX models augment this benchmark with biweekly Google Trends indicators in order to assess whether search activity improves the nowcast relative to using unemployment alone and to compare the information content of early versus late within-month search signals. Third, an unrestricted MIDAS regression incorporates weekly Google Trends series directly into the monthly equation, allowing the mixed-frequency data to be exploited without temporal aggregation. The formal specification of each model is given below.

5.1 AR(1)

Since unemployment was found to be non-stationary in levels (Table 2) but stationary in first differences, the nowcasting target variable is the monthly change in unemployment,

ΔU_t the corresponding AR(1) benchmark model is given by

$$\Delta U_t = \phi \Delta U_{t-1} + \epsilon_t \tag{3}$$

This specification models the monthly change in unemployment as a function of its own lag and serves as a simple benchmark that relies solely on past unemployment information.

5.2 ARX

To evaluate whether Google search activity contains information beyond past unemployment, I augment the AR(1) benchmark with biweekly Google Trends indicators. The resulting ARX model is

$$\Delta U_t = \alpha + \phi\, \Delta U_{t-1} + \sum_{i=1}^{K} \sum_{\ell=0}^{L_i} \beta_{i,\ell}\, \Delta G_{i,t-\ell} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \varepsilon_t, \tag{4}$$

where $\Delta G_{i,t-\ell}$ denotes the lagged changes of the i-th Google Trends series and $D_{m,t}$ are monthly dummy variables capturing residual seasonality in the non-seasonally adjusted unemployment data.

In this specification, the monthly change in unemployment, ΔU_t, depends on its own lag, on lagged changes in the high-frequency search indicators, and on seasonal controls. The term ΔU_{t-1} captures persistence in unemployment changes, while the biweekly Google Trends terms allow the model to exploit information arriving within the month.

Since unemployment in levels is integrated of order one, all models are estimated in first differences. Working with ΔU_t and $\Delta G_{i,t}$ ensures stationarity of the regression and avoids spurious relationships, while the AR(1) structure provides a parsimonious benchmark for monthly unemployment dynamics.

5.3 MIDAS

Mixed-data sampling (MIDAS) regressions, originally introduced by Ghysels et al. (2004, 2006), provide a flexible econometric framework for modelling relationships between variables observed at different sampling frequencies. A central motivation for MIDAS arises in situations where a low-frequency target series (e.g., monthly or quarterly macroeconomic indicators) is potentially explained by high-frequency predictors (such as daily financial variables, weekly surveys, or—in the present case—weekly Google search activity). MIDAS regressions allow the high-frequency information to enter directly into the low-frequency equation without requiring temporal aggregation, interpolation, or large state-space systems. This property has made MIDAS models widely used in empirical finance (e.g., volatility forecasting, return predictability) and in macroeconomic nowcasting, where timely indicators are essential.

The original MIDAS formulations proposed rely on parametric weighting schemes, most prominently the Beta and Almon lag polynomials, to summarise a potentially large number of high-frequency lags in a parsimonious way. These so-called restricted MIDAS models impose a functional form on how weekly or daily lags of the predictor affect the low-frequency outcome. The weighting function is governed by only a few shape parameters, which keeps estimation tractable even when the frequency ratio between the series is large. Other extensions of the MIDAS framework, such as transfer-function MIDAS (TF-MIDAS), which incorporates ARMA-type dynamics in the weighting process, have been proposed in the literature (Bonino-Gayoso and Garcia-Hiernaux, 2021), but these approaches typically involve more complex estimation procedures and are therefore beyond the scope of the present analysis.

A later development is the unrestricted MIDAS (U-MIDAS) model, in which no parametric weighting structure is imposed and each high-frequency lag receives its own coefficient. This variant, discussed for example in Costa et al. (2024), is estimated via ordinary least squares and avoids potential misspecification stemming from an incorrect weighting function. U-MIDAS is therefore attractive whenever full flexibility in capturing the contribution of individual weekly lags is preferred, or when the number of available high-frequency observations is sufficiently modest to allow estimation without overparameterisation.

In the context of this analysis, the unrestricted specification is particularly suitable for two reasons. First, the frequency ratio between weekly Google Trends and monthly unemployment is relatively small, making an unrestricted lag structure feasible. Second, my primary interest lies in assessing whether weekly search activity contains predictive power for unemployment, rather than in estimating a particular weighting function. I therefore adopt the U-MIDAS regression as the mixed-frequency counterpart to the biweekly ARX models introduced above.

The unrestricted MIDAS (U-MIDAS) regression allows each weekly lag of the high-frequency Google Trends series to enter the monthly unemployment equation with its own coefficient. The model is specified as

$$\Delta U_t = \alpha + \sum_{k=0}^{K} \beta_k \, \Delta G_{t-k/m} + \sum_{m=1}^{11} \gamma_m D_{m,t} + \varepsilon_t, \tag{5}$$

where $\Delta G_{t-k/m}$ denotes the k-th weekly lag of the differenced Google Trends indicator and $D_{m,t}$ are monthly dummy variables controlling for residual seasonality. Unlike the restricted MIDAS models of Ghysels et al. (2004, 2006), the U-MIDAS specification imposes no parametric weighting scheme on the high-frequency lags. This has the important advantage that the model can be estimated by ordinary least squares (OLS), as discussed in Foroni et al. (2013). For the present application, OLS estimation is particularly attractive because it avoids the nonlinear optimisation required in restricted MIDAS models, facilitates repeated estimation in a rolling-window nowcasting framework, and allows the contribution of individual weekly lags to be determined directly from the data without imposing a specific functional form.

6 Methodology

This section outlines the empirical strategy used to assess whether biweekly and weekly Google search indicators enhance the real-time prediction of monthly German unemployment. All models introduced in Section Econometric models are estimated in a pseudo-real-time environment that replicates the information constraints faced by a forecaster, and their predictive performance is evaluated over a common out-of-sample period.

6.1 Real-time information sets and forecasting design

At the end of month t, the forecaster observes the unemployment rate only up to $t-1$, because the figure for month t is released with a publication lag despite being measured in the middle of the month. In contrast, the weekly Google Trends indicators are available almost immediately and can therefore be used to form beliefs about the current state of the labour market. Letting $\Delta G_s^{(w)}$ denote the weekly change of a given search indicator, the real-time information set available at the end of month t is

$$\mathcal{I}_t = \{\Delta U_1, \ldots, \Delta U_{t-1};\ \Delta G_1^{(w)}, \ldots, \Delta G_t^{(w)}\}. \tag{6}$$

Each nowcast for month t is generated using only the information contained in $\mathcal{I}_t$. Since unemployment in levels is non-stationary, all models are estimated in first differences to ensure that the forecasting equations are well-behaved. The resulting prediction for the monthly change in unemployment, $\widehat{\Delta U_t}$, is then converted into a level forecast via

$$\widehat{U}_t = U_{t-1} + \widehat{\Delta U_t}. \tag{7}$$

6.2 Rolling estimation window

The models are estimated using a rolling-window framework, which mirrors the updating process of a real-time forecaster. For each prediction date t, the estimation sample consists of all observations strictly prior to t. The procedure begins once a sufficiently long history of data is available; in the present application, a minimum of 60 monthly observations is required so that lagged terms and seasonal dummies are reliably identified. At each step, the model parameters are re-estimated using the expanding information set, and a one-step-ahead nowcast for month t is produced.

6.3 Evaluation sample and forecast horizon

The evaluation period spans January 2019 to January 2023, reflecting both the availability of weekly Google Trends data and the publication lag of the unemployment series. For every month in this interval, a real-time nowcast $\widehat{U}_t$ is produced and compared with the subsequently released unemployment figure U_t. The resulting sequence of forecast errors,

$$e_t = \widehat{U}_t - U_t, \tag{8}$$

provides the basis for comparing the predictive performance of the competing models.

6.4 Forecast evaulation models

Forecast accuracy is assessed using root-mean-squared and mean absolute forecast errors:

$$\text{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\hat{U}_t - U_t)^2}, \quad \text{MAE} = \frac{1}{T}\sum_{t=1}^{T}|\hat{U}_t - U_t| \tag{9}$$

These metrics quantify how well each model predicts the actual unemployment level.

6.5 Statistical comparison of predictive accuracy

Pairwise comparisons of predictive accuracy across models use the Diebold–Mariano (DM) test with squared error loss. The null hypothesis of the DM test is equal predictive accuracy:

$$H_0 : E[d_t] = 0, \quad d_t = e_{1,t}^2 - e_{2,t}^2. \tag{10}$$

Rejections indicate statistically significant improvements of one model over another.

7 Tables

7.1 Google Trend variables

Table 1: Comparison of Google Trends variables: Askitas & Zimmermann (2009) vs. this study

Category	**Askitas & Zimmermann (2009)**	**This seminar (2011–2023)**
Administrative contact	*arbeitsamt*, *arbeitsagentur*	*arbeitsamt*
Interest in unemployment news	*arbeitslosenquote*	— (excluded)
Personnel consulting	*personalberater*, *personalberatung*	— (excluded)
Job board usage	*stepstone*, *jobworld*, *jobscout*, *meinestadt*, *monster*, *jobbörse*	*jobbörse*, *stepstone*, *indeed*
Application activity	—	*bewerbung*

7.2 Stationarity results

7.3 Table DF levels

Table 2: Augmented Dickey–Fuller tests in levels (monthly data, 2011M6–2023M1)

Series	Det. spec	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%
stepstone_W12	c	13	-1.6889	-2.8848	0.4368	No
indeed_W12	ct	13	-2.2393	-3.4459	0.4678	No
jobbörse_W12	ct	12	-3.1492	-3.4456	0.0950	No
arbeitsamt_W12	ct	11	-1.7611	-3.4454	0.7232	No
bewerbung_W12	ct	12	-2.0718	-3.4456	0.5619	No
stepstone_W34prev	c	13	-1.4172	-2.8848	0.5740	No
indeed_W34prev	ct	13	-2.0057	-3.4459	0.5983	No
jobbörse_W34prev	ct	12	-3.0425	-3.4456	0.1206	No
arbeitsamt_W34prev	ct	13	-2.3105	-3.4459	0.4282	No
bewerbung_W34prev	ct	12	-1.9437	-3.4456	0.6317	No
Unemp	c	13	-1.7890	-2.8848	0.3859	No

7.4 First differences tables

Table 3: Augmented Dickey–Fuller tests on first differences

Series	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone_W12	11	-3.2242	-1.9433	0.0013	Yes
indeed_W12	12	-2.4965	-1.9433	0.0121	Yes
jobbörse_W12	11	-3.0608	-1.9433	0.0022	Yes
arbeitsamt_W12	10	-5.5334	-1.9433	0.0000	Yes
bewerbung_W12	12	-3.4875	-1.9433	0.0005	Yes
stepstone_W34prev	11	-4.0136	-1.9433	0.0001	Yes
indeed_W34prev	12	-2.2670	-1.9433	0.0225	Yes
jobbörse_W34prev	11	-2.9936	-1.9433	0.0027	Yes
arbeitsamt_W34prev	12	-3.6222	-1.9433	0.0003	Yes
bewerbung_W34prev	11	-5.1217	-1.9433	0.0000	Yes
Unemp	13	-2.8170	-1.9433	0.0047	Yes

7.5 ADF for levels normal series

Table 4: Augmented Dickey–Fuller tests in levels (weekly aggregated data)

Series	Det. spec	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone	c	12	-2.3279	-2.8664	0.1631	No
indeed	ct	13	-2.5549	-3.4178	0.3011	No
jobbörse	ct	2	-6.5105	-3.4177	0.0000	Yes
arbeitsamt	ct	0	-9.8365	-3.4177	0.0000	Yes
bewerbung	ct	2	-9.1825	-3.4177	0.0000	Yes

7.6 ADF first differences MIDAS

Table 5: ADF tests on first differences of weekly indicators (no deterministic terms)

Series	Lags	ADF statistic	5% crit. value	p-value	Reject H_0 at 5%?
stepstone_weekly_diff1	12	-10.1611	-1.9415	0.0000	Yes
indeed_weekly_diff1	12	-10.8664	-1.9415	0.0000	Yes

7.7 cointegration results

Table 6: Engle–Granger ADF tests for cointegration (residual-based ADF, Case 2 critical values)

Series	ADF statistic	5% crit. value	p-value	Lags	Obs	Reject H_0 at 5%?
stepstone_W12	-3.6274	-3.4459	0.0276	13	126	Yes
indeed_W12	-3.0409	-3.4456	0.1210	12	127	No
jobbörse_W12	-2.7996	-3.4456	0.1971	12	127	No
arbeitsamt_W12	-3.1826	-3.4459	0.0879	13	126	No
bewerbung_W12	-2.3753	-3.4456	0.3928	12	127	No
stepstone_W34prev	-3.2715	-3.4456	0.0711	12	127	No
indeed_W34prev	-3.6061	-3.4459	0.0293	13	126	Yes
jobbörse_W34prev	-3.1639	-3.4456	0.0918	12	127	No
arbeitsamt_W34prev	-3.0760	-3.4459	0.1121	13	126	No
bewerbung_W34prev	-2.6002	-3.4456	0.2798	12	127	No

8 Figures

8.1 GT variables in W12 vs W34 prev

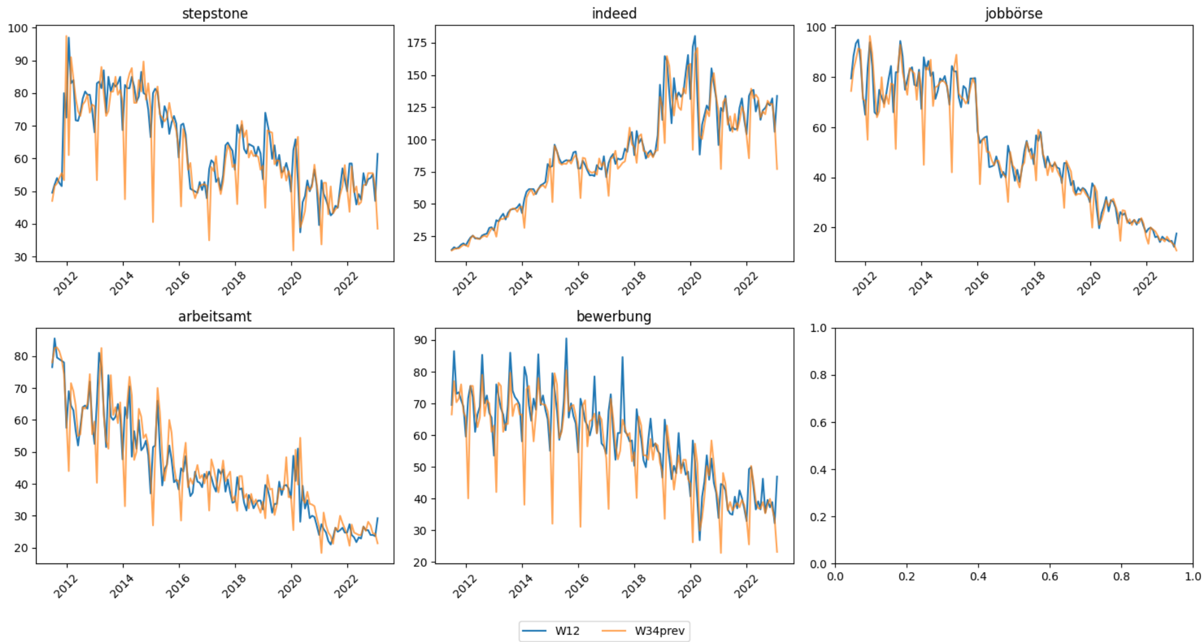

Figure 1: Biweekly Google search indicators for all keywords: W12 (blue) and W34prev (orange), 2011–2023.

8.2 Seasonal Boxplot w34prev

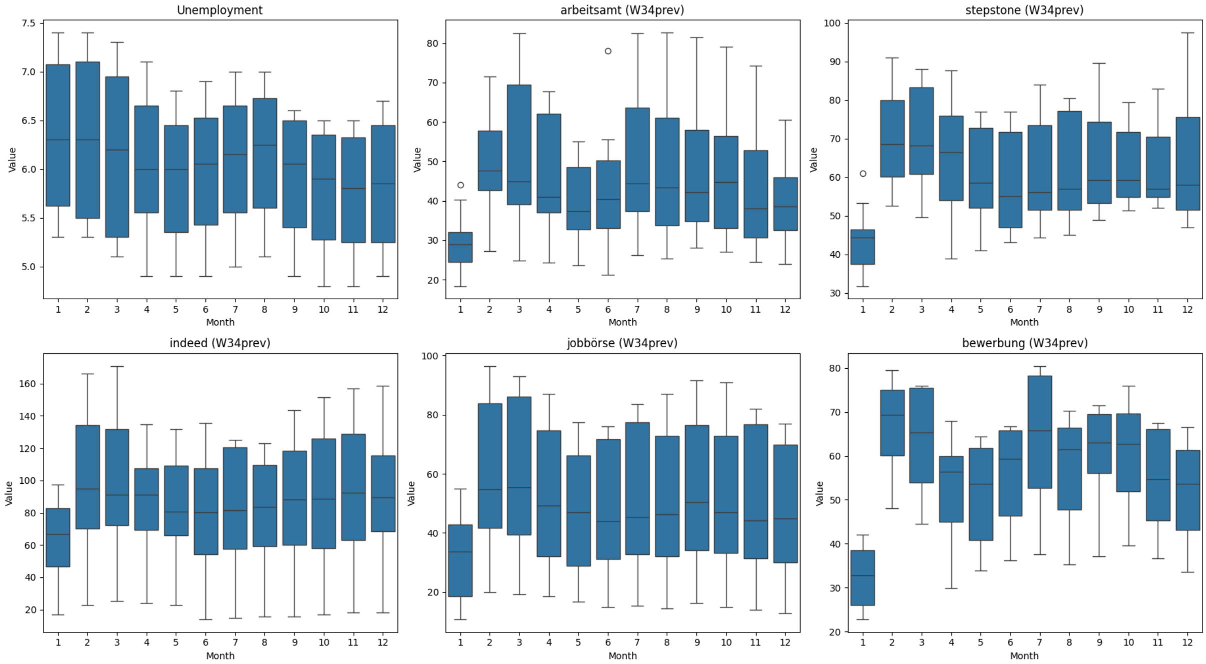

Figure 2: Seasonal boxplots for W34prev indicators.

8.3 seasonal boxplot w12

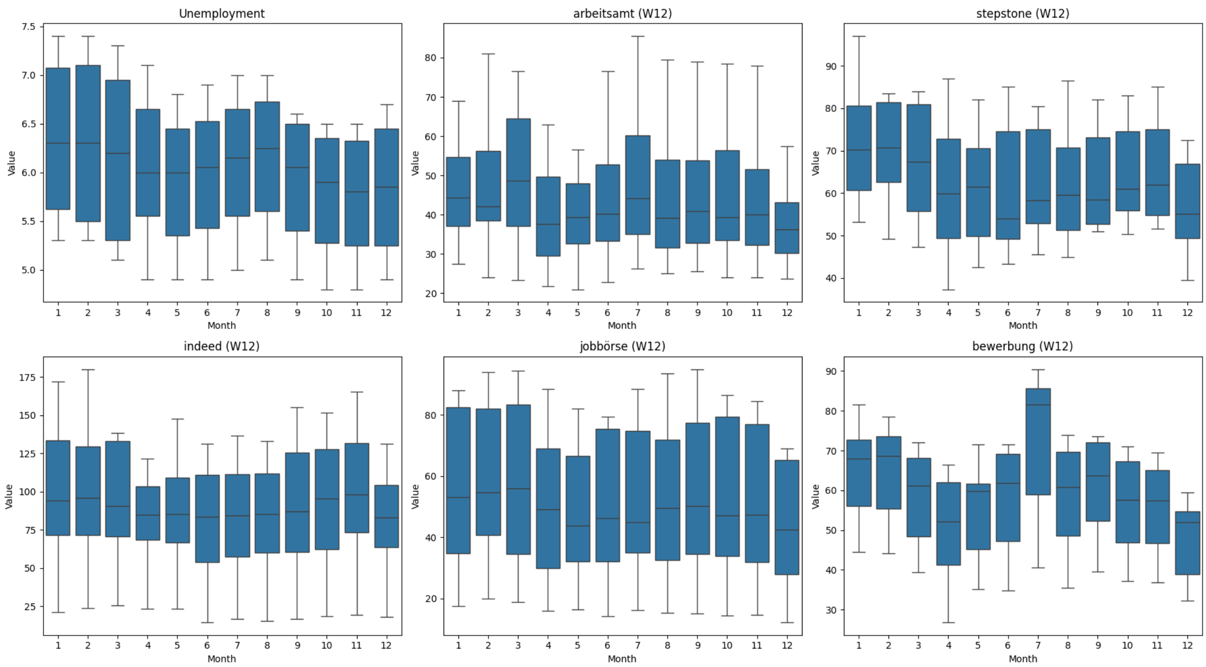

Figure 3: Seasonal boxplots for W12 indicators.

8.4 residuals cointegration

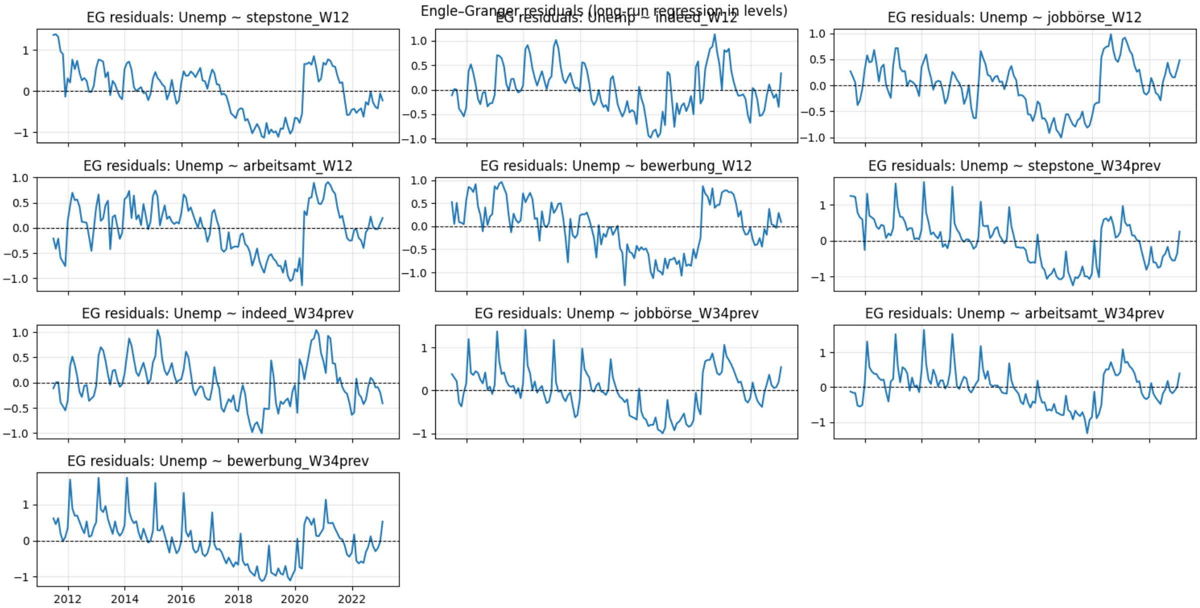

Figure 4: Residuals for cointegration

9 Appendix

9.1 Calendar example for the biweekly assignment

This appendix provides a concrete example of how weekly (Sunday) observations are assigned to the biweekly aggregates $\mathrm{W12}_M$ and $\mathrm{W34}_{M-1}$ using the "whole-week" rule described in the main text. The example avoids the three-Sundays edge case.

Consider $M =$ June 2022. The relevant Sundays around this month are:

$$\{\,\text{2022-05-15},\ \text{2022-05-22},\ \text{2022-05-29},\ \text{2022-06-05},\ \text{2022-06-12},\ \text{2022-06-19},\ \text{2022-06-26}\,\}.$$

The assignment follows the definitions in Eqs. (**??**)–(**??**):

$$\mathcal{W}^{12}_{\text{Jun22}} = \{\,\text{2022-06-05},\ \text{2022-06-12}\,\}, \qquad \mathcal{W}^{34,\text{prev}}_{\text{Jun22}} = \{\,\text{2022-05-22},\ \text{2022-05-29}\,\}.$$

Here, Sundays in the first half of June (days ≤ 15) enter $\mathrm{W12}_{\text{Jun22}}$. Sundays after the 15th of May (the second half of $M-1$) enter $\mathrm{W34}_{\text{May22}}$, which is aligned forward to June for prediction.

For a stitched weekly index $\{\tilde{x}_t\}$, the biweekly aggregates used in Eq. (**??**) evaluate to simple two-point averages:

$$\begin{aligned} X^{12}_{\text{Jun22}} &= \frac{1}{2}\Big(\tilde{x}_{\text{2022-06-05}} + \tilde{x}_{\text{2022-06-12}}\Big), \\ X^{34,\text{prev}}_{\text{Jun22}} &= \frac{1}{2}\Big(\tilde{x}_{\text{2022-05-22}} + \tilde{x}_{\text{2022-05-29}}\Big). \end{aligned}$$

For completeness, the Sundays 2022-06-19 and 2022-06-26 belong to the *second* half of June and therefore contribute to $\mathrm{W34}_{\text{Jun22}}$, which is aligned to July ($M+1$) in the main biweekly panel.

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